THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

(KATAVI, MBEYA, NJOMBE, RUKWA, RUVUMA AND SONGWE)

FORM FOUR ZONAL MOCK II EXAMINATION

031/2A PHYSICS 2A

(Actual Practical)

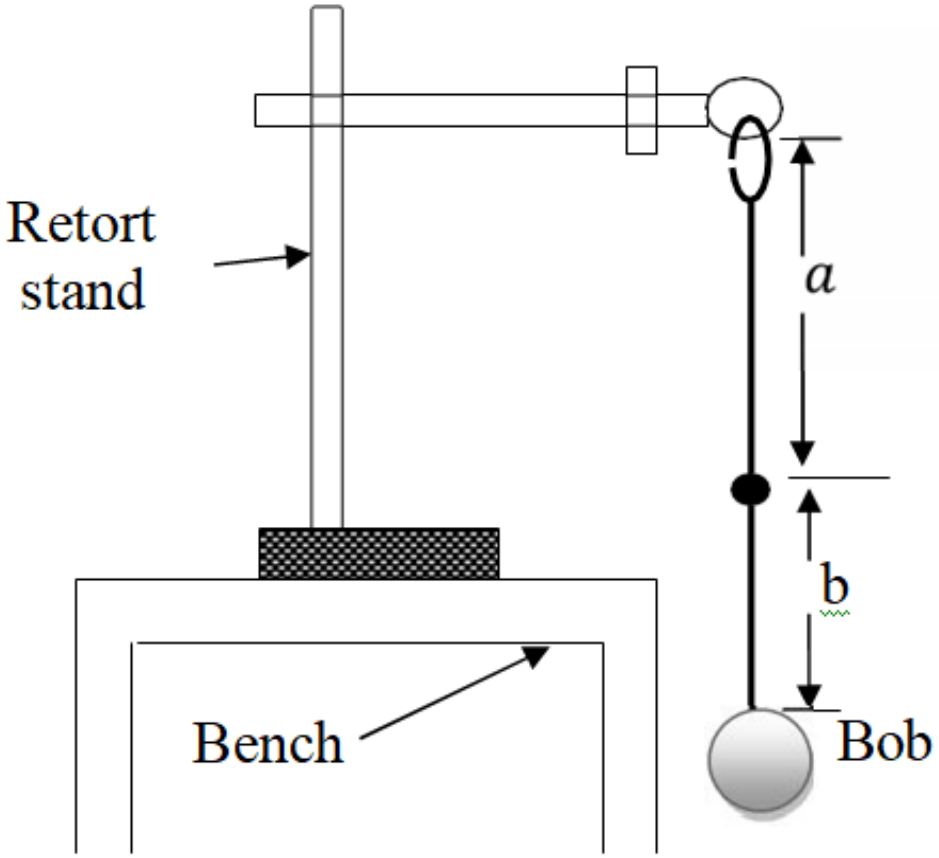
TIME: 2:30 HOURS YEAR: 2025

INSTRUCTIONS

1. This paper consists of **two (2)** questions.
2. Answer **all** questions.
3. Each question carries **twenty-five (25)** marks
4. Non programming calculators may be used.
5. All writings must be in **blue** or **black** ink, **except** drawing which must be in pencil.
6. All communication devices and any unauthorized materials are **not** allowed in the examination room.
7. Write your **Examination Number** on every page of your answer sheets provided.

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1. You are provided the following apparatus; retort stand, stop watch, cotton thread, pendulum bob, cork pads, meter rule, arrange the apparatus as shown bellow



(a) Tie a thread to a given pendulum bob, measure the length 'b' by using a given object X from the pendulum bob and mark it N

(b) Measure a distance a = 20cm from point N to the point of suspension of the pendulum bob, displace the bob to a small distance and release it so that it performs to and fro motion. Determine the time t, for 20 complete oscillations.

(c) Repeat the procedures in 1 (b) above for the value of a=40cm, 60cm, 80cm and 100 cm

(d) Tabulate the values of b, t, T, and T^2

(e) Plot a graph of T^2 (S^2) against b (cm)

(f) Find the slope, S of the graph

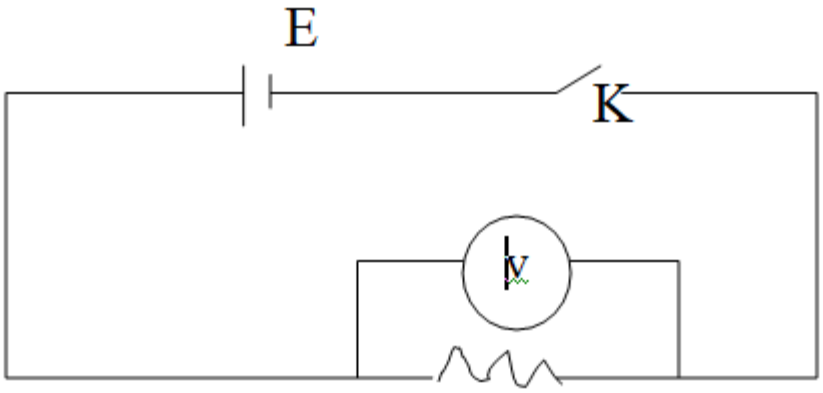
(g) Determine the value of 'b' from the equation; T^2 =S(a+b)

(h) Explain the meaning of 'b'

(i) Write two sources of error

2. You are provided with the following 2 dry cells, 8 pieces of connecting wires, voltmeter, resistance box and a key.

(a) Set up the electric circuit as shown below



(b) Starting with a resistor R = 1Ω in the resistance box, and the key closed, read and record the value of V on the voltmeter

(c) Repeat the procedure in (b) above for values of R equals to, 2Ω, 3Ω, 4Ω and 5Ω

(d) Tabulate the result including the values of 1/V

(e) Plot the graph of 1/V against R

(f) Find the slope 'S' of the graph.

(g) Record 1/V intercept as P and R intercept as Q

(h) Determine the value of the ratio: P/Q and state how the value obtained is related to the slope S of the graph